How to Use This Guide

Meteorology: An Educator's Resource for Inquiry-Based Learning for Grades 5-9 is written as a supplement to existing Earth and space science curricula for grades 5-9. The guide may be used in both formal and informal educational settings as well as at home. It should be used in conjunction with lectures, discussions, textbooks and other teaching material. This guide is not intended to be a complete course in meteorology; rather, its function is to assist educators in instilling excitement in learning about meteorology by permitting the learner to take increasing responsibility for his/her learning. The learner should experience "how we arrive at what we know," rather than memorizing what we know. This publication was developed to enhance the understanding of inquiry-based learning from the educator/teacher's perspective as well as from the learner's perspective. Inquiry-based learning has many levels. In general, inexperienced learners and younger learners will require more guidance than more-experienced and older learners who are better equipped to take responsibility for their learning. There are four levels of inquiry defined in this publication, confirmation-verification, structured inquiry, guided inquiry and open inquiry. The levels will be further defined and explained in the introductory chapter.

The guide is structured to include a short review of some principles of meteorology and facts so that they may be readily available to the educator. The Weather and Climate chapter (Chapter 2) is not intended to be used as an all-inclusive textbook, but rather an educator's guide to some of the phenomena explored in this publication. Many activities offered in this guide build upon each other and use the inquiry in the previous activity to assist in the activity that follows. Thus, this publication enhances the understanding of meteorology by beginning with basic and essential parameters of weather and then moving through mind-engaging interactions with complex meteorological systems. The "Think About This!," "Probing Further," and "Examining Results" sections are provided as examples to the educators; they may be used to stimulate the students to organize their thoughts in a particular direction. Educators may use their own creativity in stimulating student inquiry. Further educator information concerning these sections can be found in Appendix I: Suggestions For Maximizing The Use of Learner-Designed Activities.

The learner is encouraged to build and/or test a variety of weather instruments to better understand the basic factors involved in weather phenomena. The weather instruments are then brought together to form a weather station. Collecting weather information combined with existing information about cloud systems allows the learner to apply the knowledge to predict weather systems. Supplementary information and activities, which are not inquiry-based, but deemed useful by the authors, are included in the appendices, including career information Web sites in Appendix V.

An interactive video game, entitled "The Hurricane Hunters," is the second part of this project. Nightlight Studios and the authors of this guide developed the game, which should be the culminating experience in learning about meteorology as the learner has the opportunity to better understand the dynamics of hurricanes.